Bachelor of Computer Applications

Semester – V

Business Analytics

Experiment No.: 4

Title: Data Cleaning and Preparation using Power Query

Submitted By:

Name: Risha Madhuri K

Reg. No.: 2411021240063

Date: 09-09-2026

Submitted To:

Faculty Name: Mr. Aman Kumar Sharma

Department of Computer Application
Alliance University
Chandapura - Anekal Main Road, Anekal
Bengaluru – 562106

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Data Cleaning and Preparation using Power Query

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Problem Statement

A company (cafe) has received customer transaction data that contains missing values, duplicate records, and inconsistent entries such as invalid placeholder text (“ERROR”, “UNKNOWN”) and blank fields across key columns; this makes the raw dataset unreliable for reporting and analysis. Manually inspecting and correcting such issues in a spreadsheet is time-consuming, inconsistent, and error-prone as the volume of transactional data grows. This creates the need for a structured, repeatable data-cleaning process using Power Query that can detect data-quality issues, standardize formats, remove duplicates, handle missing and invalid values, and prepare a clean, analysis-ready dataset.

Methodology / Steps Followed

Step 1: Data Collection

- Dataset sourced from Kaggle: “Cafe Sales – Dirty Data for Cleaning Training” (dirty_cafe_sales.csv)
- The raw file contains 8 columns – Transaction ID, Item, Quantity, Price Per Unit, Total Spent, Payment Method, Location, and Transaction Date – across 1,000+ rows
- The dataset is intentionally “dirty,” containing missing values, invalid placeholder text (“ERROR”, “UNKNOWN”), and inconsistent entries to simulate real-world raw data.

Step 2: Data Import

- Imported dirty_cafe_sales.csv into Power BI Desktop using the Get Data → Text/CSV connector
- Set File Origin to “1252: Western European (Windows)” and Delimiter to “Comma” during import
- Data Type Detection was left as “Based on first 200 rows” at the import preview stage
- Verified the column headers and row count in the import preview before selecting Transform Data to open Power Query Editor

Step 3: Data Profiling

- Enabled Column Quality, Column Distribution, and Column Profile under the View tab in Power Query Editor to assess the health of the data
- The Column Quality report flagged uneven data quality across fields – Transaction ID was 100% valid, while Item (≈ 90% valid, 10% empty), Payment Method (≈ 68% valid, 32% empty), and Location (≈ 60% valid, 40% empty) showed significant gaps
- Quantity, Price Per Unit, and Total Spent were mostly valid (≈ 99%) but still contained isolated errors and empty cells
- Identified "ERROR" and "UNKNOWN" as recurring invalid placeholder values embedded across the Item, Payment Method, Location, Quantity, Price Per Unit, and Transaction Date columns

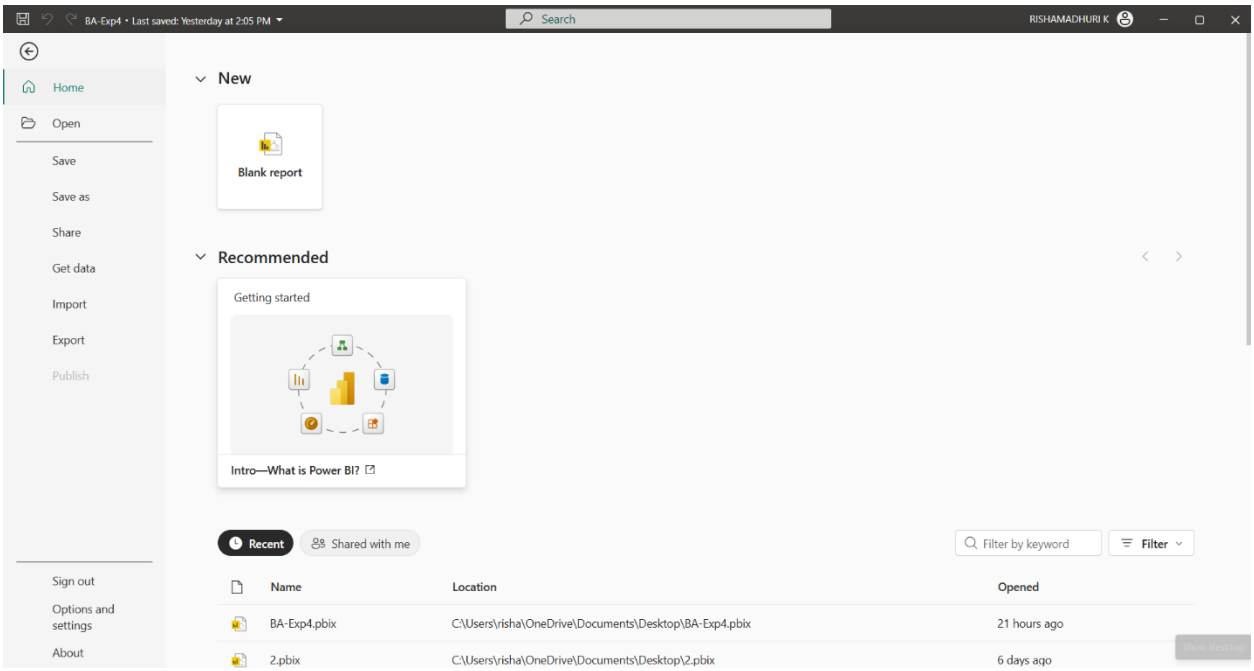
Step 4: Data Cleaning and Transformation

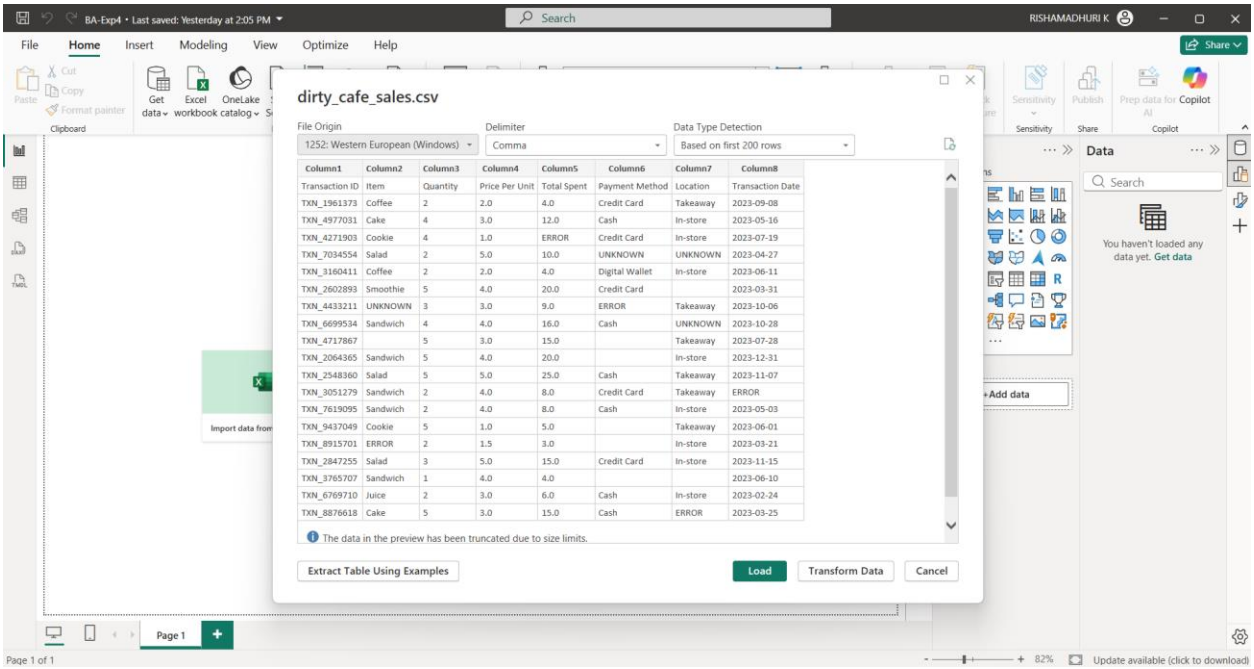
- Replaced invalid placeholder text ("ERROR", "UNKNOWN") with null across all affected columns using Replace Values
- Reviewed and handled missing values in Item, Payment Method, and Location – blank entries were either filled, categorized as "Not Specified," or removed depending on their impact on analysis
- Corrected column data types: Quantity and Price Per Unit set to Decimal/Whole Number, Transaction Date set to Date, and Transaction ID, Item, Payment Method, and Location set to Text
- Recalculated a clean Total Spent column as Quantity × Price Per Unit to fix rows where the original Total Spent value was an error or missing, then removed the original erroneous column and renamed the corrected column back to "Total Spent" (applied step: Table.RenameColumns(#"Removed Columns1", {{"Clean Total Spent", "Total Spent"}}))
- Removed duplicate rows based on Transaction ID
- Standardized the Transaction Date format consistently across all rows
- Trimmed and cleaned text fields to remove extra spaces and inconsistent casing

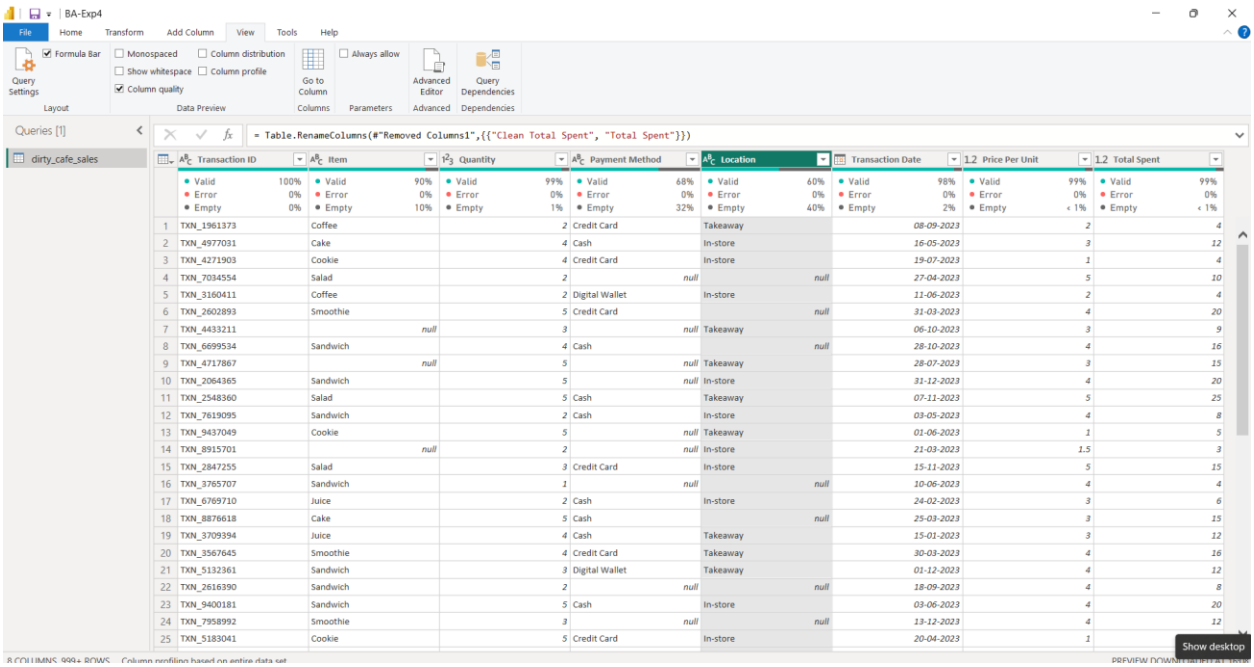
Step 5: Load and Validation

- Re-checked the Column Quality panel after cleaning to confirm a reduced Error/Empty percentage and a higher Valid percentage across all columns
- Loaded the cleaned Sales table into the Power BI data model, ready for further analysis and visualization

Output


BA-Exp4 • Last saved: Yesterday at 2:05 PM
Search
RISHAMADHURI K
Home
Open
Save
Save as
Share
Get data
Import
Export
Publish
Sign out
Options and settings
About
New
Blank report
Recommended
Getting started
Intro—What is Power BI?
Recent
Shared with me
Filter by keyword
Filter
Name
Location
Opened
BA-Exp4.pbix
C:\Users\risha\OneDrive\Documents\Desktop\BA-Exp4.pbix
21 hours ago
2.pbix
C:\Users\risha\OneDrive\Documents\Desktop\2.pbix
6 days ago


BA-Exp4 • Last saved: Yesterday at 2:05 PM
Search
RISHAMADHURI K
File
Home
Insert
Modeling
View
Optimize
Help
Share
dirty_cafe_sales.csv
File Origin
1252: Western European (Windows)
Delimiter
Comma
Data Type Detection
Based on first 200 rows
Column1 Column2 Column3 Column4 Column5 Column6 Column7 Column8
Transaction ID Item Quantity Price Per Unit Total Spent Payment Method Location Transaction Date
TXN_1961373 Coffee 2 2.0 4.0 Credit Card Takeaway 2023-09-08
TXN_4977031 Cake 4 3.0 12.0 Cash In-store 2023-05-16
TXN_4271903 Cookie 4 1.0 ERROR Credit Card In-store 2023-07-19
TXN_7034554 Salad 2 5.0 10.0 UNKNOWN UNKNOWN 2023-04-27
TXN_3160411 Coffee 2 2.0 4.0 Digital Wallet In-store 2023-06-11
TXN_2602893 Smoothie 5 4.0 20.0 Credit Card 2023-03-31
TXN_4433211 UNKNOWN 3 3.0 9.0 ERROR Takeaway 2023-10-06
TXN_6699534 Sandwich 4 4.0 16.0 Cash UNKNOWN 2023-10-28
TXN_4717867 5 3.0 15.0 Takeaway 2023-07-28
TXN_2064365 Sandwich 5 4.0 20.0 In-store 2023-12-31
TXN_2548360 Salad 5 5.0 25.0 Cash Takeaway 2023-11-07
TXN_3051279 Sandwich 2 4.0 8.0 Credit Card Takeaway ERROR
TXN_7619095 Sandwich 2 4.0 8.0 Cash In-store 2023-05-03
TXN_9437049 Cookie 5 1.0 5.0 Takeaway 2023-06-01
TXN_8915701 ERROR 2 1.5 3.0 In-store 2023-03-21
TXN_2847255 Salad 3 5.0 15.0 Credit Card In-store 2023-11-15
TXN_3765707 Sandwich 1 4.0 4.0 2023-06-10
TXN_6769710 Juice 2 3.0 6.0 Cash In-store 2023-02-24
TXN_8876618 Cake 5 3.0 15.0 Cash ERROR 2023-03-25
The data in the preview has been truncated due to size limits.
Extract Table Using Examples
Load
Transform Data
Cancel
Data
Search
You haven't loaded any data yet. Get data
Add data
Page 1
Page 1 of 1
82%
Update available (click to download)



- Power Query Editor view of the dirty_cafe_sales query showing the 8 source columns (Transaction ID, Item, Quantity, Payment Method, Location, Transaction Date, Price Per Unit, Total Spent) with Column Quality, Column Distribution, and Column Profile indicators enabled
- Column Quality statistics displayed per column (Valid / Error / Empty percentages), highlighting Payment Method and Location as the columns with the highest proportion of missing data
- Get Data preview of dirty_cafe_sales.csv showing the original raw values, including “ERROR” and “UNKNOWN” placeholders and blank cells, before cleaning was applied
- Applied Steps pane recording the cleaning transformations, including the column removal and rename step used to restore a corrected “Total Spent” column

Conclusion

This experiment successfully used Power Query to transform a raw, error-prone cafe sales dataset into a clean, consistent, and analysis-ready table. By systematically profiling column quality, replacing invalid placeholder values, handling missing entries, correcting data types, recalculating derived fields, and removing duplicates, the reliability of the dataset was significantly improved. The exercise demonstrated how Power Query's profiling and transformation tools can replace slow, manual, spreadsheet-based cleaning with a repeatable, auditable process suitable for downstream reporting and analysis.

Learning Outcomes

- Learned to use Power Query's Column Quality, Column Distribution, and Column Profile tools to diagnose data-quality issues before cleaning
- Gained practical experience identifying and replacing invalid placeholder values ("ERROR", "UNKNOWN") and handling missing/null values
- Understood how to correct column data types and recalculate derived columns to fix corrupted numeric fields
- Practiced removing duplicate records and standardizing inconsistent date and text formats
- Learned to document and trace transformation logic through Power Query's Applied Steps for reproducibility

Business Insights

- A large share of missing data was concentrated in Payment Method and Location, suggesting these fields may need better validation at the point of sale to avoid analysis gaps
- Placeholder values like "ERROR" and "UNKNOWN" indicate upstream data-entry or system-integration issues that should be fixed at the source rather than repeatedly cleaned downstream
- Recalculating Total Spent from Quantity and Price Per Unit exposed rows where the original recorded revenue figure was unreliable, which could otherwise have understated or overstated sales performance
- A clean, validated dataset is a prerequisite for any further sales, revenue, or product-performance analysis, reinforcing the value of a dedicated data-cleaning stage before dashboard development